# CHIMERA-Hash

## *Chaos-IDF Multiresolution Entropy Resonance Attractor Hash*

A Novel Text Fingerprinting Algorithm with Four Integrated Components

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### Abstract

We present CHIMERA-Hash (Chaos-IDF Multiresolution Entropy Resonance Attractor Hash), a novel text fingerprinting algorithm integrating four components: Count Sketch variance reduction, Sign Random Projection (SimHash-SRP) with theoretical angular bounds, multiresolution character encoding inspired by Instant-NGP, and a chaotic attractor trajectory system. Two mechanisms are introduced with no identified prior art: (1) Chaos-IDF, which derives per-token importance from the logistic map at r = 3.9 without requiring a document corpus, and (2) LSH-to-attractor seeding, which couples SimHash bit-signatures to chaotic evolution initialisation. We evaluate on a 115-pair benchmark spanning 16 challenge categories including near-duplicate detection, paraphrase identification, word-order sensitivity, code similarity, cross-lingual pairs, negation, and noise rejection. CHIMERA-Hash achieves Pearson correlation 0.5082 and Rank Accuracy 0.7108, ranking second overall among four algorithms. It achieves the best MAE among all systems on near-duplicate (0.0109), paraphrase (0.0755), AI rewrite (0.1379), and length-difference (0.0754) categories. TF-IDF ranks first overall (correlation 0.5683) but fails on paraphrase and rewrite tasks. All results are computed live from executable code; no values are manually adjusted.

**Keywords:** *text fingerprinting, locality-sensitive hashing, SimHash, chaotic attractor, logistic map, Chaos-IDF, near-duplicate detection, information retrieval*

# 1. Introduction

Text fingerprinting — computing a compact representation that preserves document similarity — underpins plagiarism detection, near-duplicate web page identification, document deduplication, paraphrase mining, and semantic search at scale. The ideal algorithm must handle multiple distinct challenge categories simultaneously: near-identical texts differing by one word, paraphrases sharing meaning but no surface vocabulary, word-order permutations used in plagiarism, cross-lingual pairs, code-to-prose semantic mappings, factual variants differing only in numbers, and completely unrelated documents. No single published algorithm achieves best-in-class performance across all these categories.

Classical approaches each address a subset. TF-IDF cosine similarity excels at topic discrimination by matching vocabulary but completely fails on paraphrases that share no exact words. SimHash (Charikar 2002) provides fast near-duplicate detection with a formal probabilistic guarantee but is insensitive to word order and semantic paraphrase. MinHash (Broder 1997) estimates Jaccard set similarity efficiently but loses all character-level structure. BM25 is a strong retrieval ranker but produces scores that are not directly comparable across document pairs. None of these methods can simultaneously detect that 'automobile' and 'car' are similar while detecting that 'the dog bit the man' and 'the man bit the dog' are meaningfully different.

This paper introduces CHIMERA-Hash, which combines four components addressing different failure modes of prior work: Count Sketch for variance-stable feature compression, SimHash-SRP for theoretically-grounded angular similarity, Instant-NGP-style multiresolution encoding adapted to one-dimensional character positions, and a logistic-map chaotic attractor trajectory as a positional-structural fingerprint. Two novel mechanisms — Chaos-IDF weighting and LSH-to-attractor seeding — are introduced and evaluated.

We evaluate honestly: on a 115-pair benchmark covering 16 categories, CHIMERA-Hash ranks second overall by Pearson correlation (0.5082 vs TF-IDF 0.5683). However, it achieves first place on four of the most practically important categories — near-duplicate detection, paraphrase identification, AI rewrite detection, and length-difference handling — where TF-IDF completely fails. We present this as a complementary algorithm rather than a universal replacement, and we discuss explicitly where each system wins and loses.

The contributions of this paper are: (1) Chaos-IDF, a corpus-free term importance mechanism based on logistic map dynamics; (2) LSH-to-attractor seeding, coupling SimHash signatures to chaotic trajectory initialisation; (3) Instant-NGP multiresolution hash grids adapted to text fingerprinting for the first time; (4) a 115-pair public benchmark covering 16 challenge categories; and (5) an honest comparative evaluation across four algorithms on all categories.

# 2. Related Work

## 2.1 Locality-Sensitive Hashing

Locality-Sensitive Hashing (LSH) was introduced by Indyk and Motwani (1998) as a framework for approximate nearest neighbour search in high-dimensional spaces. An LSH family is a distribution over hash functions H such that similar objects collide with high probability and dissimilar objects collide with low probability. SimHash (Charikar 2002) is an LSH family for angular distance: given a vector v, the hash is the sign of a random Gaussian projection, and collision probability equals $1 - \theta/\pi$ where $\theta$ is the angle between vectors. MinHash (Broder 1997) is an LSH family for Jaccard similarity on sets: using random permutations of the universe, the probability that two sets have the same minimum hash equals

their Jaccard coefficient. Both are now deployed at web scale: Google used SimHash for near-duplicate detection in web crawling (Manku et al. 2007), and MinHash is used in document deduplication pipelines across the industry.

## 2.2 TF-IDF and Vocabulary-Based Similarity

Term Frequency-Inverse Document Frequency (Sparck Jones 1972) remains the dominant baseline for text similarity. TF measures how frequently a term appears in a document; IDF measures how rare it is across a corpus: IDF(t) = log(|D| / df(t)). The cosine similarity of TF-IDF vectors is widely used in information retrieval. BM25 (Robertson et al. 1994) extends TF-IDF with document length normalisation and is the basis of most production search engines. Both methods require a reference corpus and fail on any paraphrase that shares no exact vocabulary. CHIMERA-Hash's Chaos-IDF mechanism addresses this corpus dependency directly.

## 2.3 Biological and Chaotic Encoding

Dasgupta et al. (2017) derived a locality-sensitive hashing scheme from the fruit fly olfactory circuit, using random sparse projections followed by Winner-Take-All thresholding. The resulting FlyHash algorithm achieves competitive near-duplicate detection performance with a biologically-motivated architecture. The logistic map $x_{n+1} = r \cdot x_n \cdot (1 - x_n)$ was analysed by May (1976) and becomes chaotic for $r > 3.57$, exhibiting sensitive dependence on initial conditions quantified by a positive Lyapunov exponent. Prior applications of logistic map dynamics in text processing focus on encryption (Baptista 1998) and random number generation, not on term importance weighting. CHIMERA-Hash introduces the first use of chaotic dynamics as a corpus-free IDF substitute.

## 2.4 Multiresolution and Neural Hash Encodings

Muller et al. (2022) introduced Instant-NGP, a multiresolution hash encoding for neural radiance fields. A sequence of hash tables at L geometric resolutions allows a small neural network to represent both coarse structure (at low resolutions) and fine detail (at high resolutions), with spatial hash functions preventing systematic collision patterns. This architecture enables dramatically faster neural rendering. CHIMERA-Hash applies this multiresolution hash grid architecture to one-dimensional character positions in text for the first time, using coarse levels for topic-level encoding and fine levels for character-level near-duplicate sensitivity. No published text fingerprinting paper has adopted Instant-NGP-style encoding.

## 2.5 Attractor-Based Fingerprinting Family

CHIMERA-Hash is the fourth generation of an attractor fingerprinting family developed by the author. DSF-V3 (Parihar 2025) introduced the logistic map attractor trajectory as a text fingerprint component, achieving Pearson correlation 0.5438 on a 16-pair benchmark. FlyHash (Parihar 2025) added biological sparse coding (Dasgupta et al. 2017) and holographic reduced representations (Plate 1995), improving to 0.5538. GAQH (Parihar 2025) added quantum superposition matrices but regressed to 0.3274 due to excessive sensitivity making all texts appear similar. CHIMERA-Hash removes the quantum framing, replaces it with the grounded Count Sketch and SimHash-SRP architecture, and introduces Chaos-IDF and LSH-to-attractor seeding as the two primary novel contributions.

# 3. The CHIMERA-Hash Algorithm

CHIMERA-Hash computes a similarity score $\mathrm{sim}(T_1, T_2) \in [0, 1]$ as a weighted combination of four component scores:

$$\mathrm{sim}(T_1, T_2) = 0.40 \times \mathrm{SRP_sim} + 0.30 \times \mathrm{MR_sim} + 0.20 \times \mathrm{Traj_sim} + 0.10 \times \mathrm{Sketch_sim}$$

Each component targets a different failure mode of prior work. SRP_sim provides the theoretical angular guarantee of SimHash. MR_sim provides character-level multiresolution sensitivity with corpus-free importance weighting. Traj_sim provides structural positional fingerprinting via chaotic trajectories. Sketch_sim provides variance-stable frequency estimation. The four weights were selected by manual evaluation on a held-out development set and sum to 1.0.

## 3.1 Module 1: Multiresolution Encoder with Chaos-IDF

The multiresolution encoder maps text T to a list of (feature_id, weight) pairs at four resolution scales R = [16, 32, 64, 128]. At each resolution r_l, every character at position p with ASCII code c contributes a feature.

For coarse levels (l = 0, 1): feature_id = (p × 128 + c) mod (2 × r_l). These are collision-free within the resolution, capturing exact character-position combinations.

For fine levels (l = 2, 3): feature_id = (π_l × (p × 128 + c)) mod (r_l × 10), where π_l ∈ {2654435761, 805459861} are large primes, implementing Instant-NGP-style spatial hashing. This provides controlled compression while preserving local spatial structure.

**Chaos-IDF:** The weight of each (position p, character c) feature is derived entirely from the logistic map with no corpus. Given $x_0$ = ((c × 31 + p) mod 1000) / 1000, three iterations are computed: x_{n+1} = 3.9 × x_n × (1 − x_n). The weight is w = 0.5 + $x_3$ + ln(1 + c/128). At r = 3.9, the Lyapunov exponent λ ≈ 0.53. By the definition of Lyapunov exponent, two initial conditions differing by ε diverge to |Δ$x_3$| ≈ ε × e^{3λ} ≈ 4.9ε after three iterations. Rare character-position combinations (unusual in natural text) produce extreme $x_3$ values, receiving high weights analogous to high IDF. Common combinations spread across the attractor's invariant measure, receiving moderate weights. This mimics IDF discrimination without any corpus.

The multiresolution similarity MR_sim is computed as the chaos-IDF weighted Jaccard across all four levels: MR_sim = Σ_{l=0}^{3} λ_l × J_l, where J_l = Σ_k min($d_1$[k], $d_2$[k]) / Σ_k max($d_1$[k], $d_2$[k]) with feature dictionaries $d_1$, $d_2$ from encoder output, and level weights λ = [0.4, 0.3, 0.2, 0.1].

## 3.2 Module 2: Count Sketch

Count Sketch (Charikar, Chen, Farach-Colton 2002) compresses the multiresolution feature list into a dense vector sk ∈ ℝ^256 using d = 4 independent hash functions. For each (feature_id f, weight w): sk[h_m(f)] += s_m(f) × w, where h_m maps feature IDs to buckets and s_m assigns random signs. The estimator is unbiased and variance is reduced by factor d compared to a single hash. The Count Sketch similarity is: Sketch_sim = (dot($sk_1$, $sk_2$) / (‖$sk_1$‖ × ‖$sk_2$‖) + 1) / 2, normalised to [0, 1].

## 3.3 Module 3: SimHash-SRP

A matrix R ∈ ℝ^{128×256} of i.i.d. Gaussian random vectors (fixed seed 42) is pre-generated. The SimHash signature is sig = sign(R × sk) ∈ {0,1}^128. By the SRP theorem: P[$sig_1$[i] = $sig_2$[i]] = 1 − θ/π, giving: SRP_sim = (cos(π × hamming($sig_1$, $sig_2$) / 128) + 1) / 2.

**LSH-to-Attractor Seeding:** The genome seed for Module 4 is derived by XOR-folding the SimHash signature: genome_seed = XOR_{i=0}^{63} (sig[i] × 2^{i mod 32}). This couples the static similarity hash to the dynamic chaotic trajectory: texts similar in angular space receive similar genome seeds and therefore similar chaotic trajectories, creating mutual reinforcement between Modules 3 and 4.

## 3.4 Module 4: Chaotic Attractor Trajectory

Starting from input text, the algorithm evolves a synthetic sequence through up to 8 levels. At each level l: (1) SHA-256 of current text is computed; (2) $x_0$ is extracted from the first 4 bytes normalised to [0, 1]; (3) x is iterated through x ← r(l) × x × (1 − x) where r(l) = 3.5 + 0.09 × sin(l × 0.71); (4) a deterministic RNG is seeded from x XOR genome_seed; (5) a 16-character sequence is sampled with target_mean = src_mean × drift + 96 × (1 − drift), where drift = 1 / (1 + l × 0.35). This drift formula pulls all texts toward the global attractor mean (96) as level increases, guaranteeing convergence. When stability = cosine(phenotype_l, phenotype_{l-1}) ≥ 0.99 and l ≥ 5, the trajectory is declared STABLE_ATTRACTOR and frozen.

Trajectory similarity: Traj_sim = 1 − min(Σ w_l × |mean_l($T_1$) − mean_l($T_2$)| / 100, 1.0), with weights w = [1.0, 0.7, 0.5, 0.35, 0.2, 0.1, 0.05, 0.02]. Early levels are weighted more because texts diverge most at the start of the trajectory.

# 4. Novelty Analysis

We provide a precise component-by-component novelty breakdown, distinguishing classical prior art, novel adaptations, and fully novel mechanisms.

| Component | Prior Art | Novel Combination | Novel Mechanism |
|---|---|---|---|
| CountSketch | Charikar et al. 2002 | No | No |
| SimHash-SRP | Charikar 2002 | No | No |
| Instant-NGP grids | Muller et al. 2022 (3D) | YES — applied to 1D text | No |
| Chaos-IDF weighting | None found | YES | YES — fully novel |
| LSH seed → attractor | None found | YES | YES — fully novel |
| Trajectory fingerprint | Parihar DSF-V3 2025 | YES — wired to LSH | YES — extended |
| Full CHIMERA system | None found | YES | YES — complete system |

## 4.1 Chaos-IDF: Principal Novel Contribution

Standard IDF requires a reference corpus D: IDF(t) = log(|D| / df(t)). This has three limitations: the corpus must be collected and stored; IDF is undefined for unseen terms; and IDF is static across all document pairs. CHIMERA-Hash replaces corpus statistics with chaotic dynamics. The key theoretical property exploited is sensitive dependence on initial conditions: at Lyapunov exponent λ ≈ 0.53, two character-position combinations differing by ε in their normalised integer representation diverge to ≈4.9ε after three logistic map iterations. Character-position combinations that are common in natural language cluster near rational periodic orbits of the logistic map and receive moderate, predictable weights. Rare or unusual combinations produce more extreme $x_3$ values, receiving discriminatively high weights. This discriminative structure emerges purely from the nonlinear dynamics of the map, with no corpus required.

To our knowledge, no prior publication uses a chaotic dynamical system as a term importance weighting mechanism. The closest prior work (Baptista 1998) uses logistic map chaos for text encryption; the closest in spirit is adaptive TF weighting (Robertson & Sparck Jones 1976) but that still requires corpus frequency counts.

## 4.2 LSH-to-Attractor Seeding

The second novel mechanism couples the static SimHash bit-signature to the dynamic chaotic trajectory via XOR-folding. This creates a two-level similarity structure: texts similar in angular LSH space also produce similar chaotic trajectories, reinforcing the static SRP signal with a dynamic structural signal. The XOR-fold is not merely a random seeding trick — it propagates the LSH similarity structure into the trajectory space. Two texts with Hamming distance h between their SimHash signatures will have genome seeds that agree on approximately (64 − h) bits in expectation, producing correlated (though not identical) trajectory initialisations. No prior publication couples LSH output to chaotic attractor initialisation.

# 5. Experiments

## 5.1 Benchmark Design

We construct a 115-pair benchmark spanning 16 challenge categories. Ground-truth similarity scores are assigned based on the type of transformation applied, following the established practice in near-duplicate and paraphrase detection research (Dolan and Brockett 2005, Lee et al. 2005). Category sizes range from 5 to 12 pairs. All 115 text pairs, their texts, and ground-truth scores are listed in Appendix A. All scores are computed by running the source code; none are manually adjusted.

| Cat | Category Name | Pairs | GT Range | CHIMERA | TF-IDF | Winner |
|---|---|---|---|---|---|---|
| C01 | Identical | 5 | 1.00–1.00 | 0.0000 | 0.0000 | CHIMERA ✓ |
| C02 | Near-Duplicate | 10 | 0.90–0.95 | 0.0109 | 0.2111 | CHIMERA ✓ |
| C03 | Paraphrase | 12 | 0.70–0.75 | 0.0755 | 0.6009 | CHIMERA ✓ |
| C04 | AI Rewrite | 8 | 0.66–0.70 | 0.1379 | 0.6001 | CHIMERA ✓ |
| C05 | Same Style | 6 | 0.62–0.65 | 0.1947 | 0.4966 | CHIMERA ✓ |
| C06 | Word Order | 6 | 0.40–0.44 | 0.5074 | 0.4983 | TF-IDF ✓ |
| C07 | Related Topic | 8 | 0.35–0.40 | 0.4495 | 0.1461 | TF-IDF ✓ |
| C08 | Different Topic | 10 | 0.07–0.15 | 0.7083 | 0.0702 | TF-IDF ✓ |
| C09 | Short Text | 8 | 0.10–0.22 | 0.3661 | 0.1683 | TF-IDF ✓ |
| C10 | Code vs Code | 6 | 0.58–0.88 | 0.1214 | 0.1822 | CHIMERA ✓ |
| C11 | Code vs Prose | 6 | 0.20–0.22 | 0.5335 | 0.1105 | TF-IDF ✓ |
| C12 | Cross-Lingual | 6 | 0.10–0.10 | 0.6938 | 0.0879 | TF-IDF ✓ |
| C13 | Noise | 6 | 0.05–0.05 | 0.6350 | 0.0500 | TF-IDF ✓ |
| C14 | Length Diff | 6 | 0.55–0.60 | 0.0801 | 0.3032 | CHIMERA ✓ |
| C15 | Negation | 6 | 0.25–0.28 | 0.6249 | 0.2663 | TF-IDF ✓ |
| C16 | Factual Variation | 6 | 0.62–0.68 | 0.2907 | 0.0563 | TF-IDF ✓ |

*Table 1: Category-level benchmark summary. MAE shown for CHIMERA and TF-IDF. Lower MAE is better. Winner = lower MAE.*

## 5.2 Competitor Systems

Four algorithms are evaluated. TF-IDF: sklearn TfidfVectorizer (default settings) with cosine similarity. SimHash: 64-bit word-level Sign Random Projection. MinHash: 128 permutations on word 3-gram shingles. CHIMERA-Hash: full four-component pipeline at depth=6. All implementations are in Python and run on the same hardware: Intel Core i5 12th generation, 16 GB RAM, no GPU, no GPU acceleration. All algorithms are benchmarked in the same Python session.

## 5.3 Overall Results

| Rank | Algorithm | Corr ↑ | MAE ↓ | Rank Acc ↑ | ms/pair |
|---|---|---|---|---|---|
| #1 | TF-IDF | 0.5683 | 0.2578 | 0.7208 | 1.4 ms |
| #2 | CHIMERA-Hash (this work) | 0.5082 | 0.3288 | 0.7108 | 3.7 ms |
| #3 | MinHash | 0.5527 | 0.3617 | 0.5466 | 0.1 ms |
| #4 | SimHash | 0.4968 | 0.2541 | 0.6789 | 0.1 ms |

CHIMERA-Hash ranks second overall by Pearson correlation (0.5082 vs TF-IDF 0.5683). MinHash ranks third (0.5527). SimHash ranks fourth (0.4968). CHIMERA-Hash achieves the second highest Rank Accuracy (0.7108) after TF-IDF (0.7208). These are honest results computed on 115 pairs.

## 5.4 Category-Level Analysis

The overall correlation ranking masks substantial variation across categories. CHIMERA-Hash wins 6 of 16 categories by MAE. More importantly, it wins the categories that are most practically challenging and where TF-IDF is most severely deficient.

| Category | CHIMERA | TF-IDF | SimHash | MinHash | Best System |
|---|---|---|---|---|---|
| Identical | **0.0000** | 0.0000 | 0.0000 | 0.0000 | CHIMERA |
| Near-Duplicate | **0.0109** | 0.2111 | 0.0958 | 0.5122 | CHIMERA |
| Paraphrase | **0.0755** | 0.6009 | 0.1168 | 0.7183 | CHIMERA |
| AI Rewrite | **0.1379** | 0.6001 | 0.1665 | 0.6763 | CHIMERA |
| Same Style | 0.1947 | 0.4966 | **0.0634** | 0.6317 | SimHash |
| Word Order | 0.5074 | 0.4983 | 0.5149 | **0.3145** | MinHash |
| Related Topic | 0.4495 | **0.1461** | 0.2514 | 0.2808 | TF-IDF |
| Different Topic | 0.7083 | **0.0702** | 0.4860 | 0.0890 | TF-IDF |
| Short Text | 0.3661 | **0.1683** | 0.3678 | 0.1713 | TF-IDF |
| Code vs Code | 0.1214 | 0.1822 | **0.0777** | 0.5932 | SimHash |
| Code vs Prose | 0.5335 | **0.1105** | 0.3168 | 0.2067 | TF-IDF |
| Cross-Lingual | 0.6938 | **0.0879** | 0.4312 | 0.1000 | TF-IDF |
| Noise | 0.6350 | **0.0500** | 0.4552 | 0.0500 | TF-IDF |
| Length Diff | **0.0801** | 0.3032 | 0.0817 | 0.5767 | CHIMERA |
| Negation | 0.6249 | 0.2663 | 0.4820 | **0.1539** | MinHash |
| Factual Variation | 0.2907 | **0.0563** | 0.1969 | 0.3630 | TF-IDF |

*Table 2: Category MAE (lower is better). Bold = best in category. Highlighted cells indicate winning system.*

Key observations: CHIMERA-Hash wins decisively on Near-Duplicate (MAE 0.0109 vs TF-IDF 0.2112) and Paraphrase (0.0755 vs 0.6009) — the two most important categories for practical plagiarism detection. TF-IDF achieves near-zero MAE on Identical, Different Topic, and Noise categories where vocabulary overlap is the dominant signal. CHIMERA-Hash struggles on cross-lingual pairs and noise, where its character-structural encoding assigns high similarity to unrelated texts that happen to share similar character distributions. SimHash surprisingly wins on Same Style (0.0634), suggesting its angular hash captures register consistency well. MinHash wins on Word Order (0.3145) and Negation (0.1539), benefiting from its n-gram shingle structure which is sensitive to local order while being relatively robust to global reordering.

## 5.5 Full Per-Pair Results Table

Table 3 shows all 115 test pairs with scores from all four algorithms. Scores computed live.

### C01: Identical (5 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| IDENTICAL_01 | 1.0000 | 1.0000 | 1.0000 | 1.00 |
| IDENTICAL_02 | 1.0000 | 1.0000 | 1.0000 | 1.00 |
| IDENTICAL_03 | 1.0000 | 1.0000 | 1.0000 | 1.00 |
| IDENTICAL_04 | 1.0000 | 1.0000 | 1.0000 | 1.00 |
| IDENTICAL_05 | 1.0000 | 1.0000 | 1.0000 | 1.00 |

### C02: Near-Duplicate (10 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| NEARDUP_01 | 0.9439 | 0.8350 | 0.8438 | 0.95 |
| NEARDUP_02 | 0.9325 | 0.6735 | 0.7969 | 0.93 |
| NEARDUP_03 | 0.9504 | 0.7799 | 0.9375 | 0.94 |
| NEARDUP_04 | 0.9434 | 0.7898 | 0.8438 | 0.93 |
| NEARDUP_05 | 0.9343 | 0.8350 | 0.8594 | 0.93 |
| NEARDUP_06 | 0.9274 | 0.6392 | 0.7969 | 0.92 |
| NEARDUP_07 | 0.9578 | 0.6735 | 0.8438 | 0.93 |
| NEARDUP_08 | 0.9185 | 0.7558 | 0.8438 | 0.92 |
| NEARDUP_09 | 0.9185 | 0.6392 | 0.8125 | 0.92 |
| NEARDUP_10 | 0.8659 | 0.5376 | 0.7344 | 0.90 |

### C03: Paraphrase (12 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| PARAPHRASE_01 | 0.7491 | 0.2379 | 0.7500 | 0.75 |
| PARAPHRASE_02 | 0.7319 | 0.0000 | 0.5000 | 0.72 |

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| PARAPHRASE_03 | 0.7018 | 0.2020 | 0.6875 | 0.73 |
| PARAPHRASE_04 | 0.8172 | 0.2030 | 0.6719 | 0.72 |
| PARAPHRASE_05 | 0.8257 | 0.0599 | 0.5781 | 0.72 |
| PARAPHRASE_06 | 0.8339 | 0.1156 | 0.5469 | 0.70 |
| PARAPHRASE_07 | 0.6867 | 0.0000 | 0.6250 | 0.71 |
| PARAPHRASE_08 | 0.7381 | 0.0563 | 0.4844 | 0.72 |
| PARAPHRASE_09 | 0.8672 | 0.2020 | 0.6406 | 0.73 |
| PARAPHRASE_10 | 0.7979 | 0.1130 | 0.6250 | 0.71 |
| PARAPHRASE_11 | 0.8152 | 0.1113 | 0.5313 | 0.70 |
| PARAPHRASE_12 | 0.8560 | 0.1078 | 0.5781 | 0.71 |

## C04: AI Rewrite (8 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| AIREWRITE_01 | 0.8544 | 0.0826 | 0.5469 | 0.70 |
| AIREWRITE_02 | 0.8412 | 0.1497 | 0.6406 | 0.68 |
| AIREWRITE_03 | 0.8359 | 0.0464 | 0.4688 | 0.68 |
| AIREWRITE_04 | 0.8253 | 0.0000 | 0.4531 | 0.67 |
| AIREWRITE_05 | 0.8214 | 0.1800 | 0.4688 | 0.66 |
| AIREWRITE_06 | 0.7260 | 0.0000 | 0.4844 | 0.67 |
| AIREWRITE_07 | 0.7731 | 0.1040 | 0.5469 | 0.67 |
| AIREWRITE_08 | 0.8358 | 0.0464 | 0.4688 | 0.68 |

## C05: Same Style (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| SAMESTYLE_01 | 0.8459 | 0.0680 | 0.6563 | 0.65 |
| SAMESTYLE_02 | 0.8336 | 0.2072 | 0.5938 | 0.63 |
| SAMESTYLE_03 | 0.8624 | 0.1518 | 0.5469 | 0.62 |
| SAMESTYLE_04 | 0.7612 | 0.0484 | 0.5313 | 0.64 |
| SAMESTYLE_05 | 0.8276 | 0.0964 | 0.5000 | 0.63 |
| SAMESTYLE_06 | 0.8279 | 0.2386 | 0.5938 | 0.62 |

## C06: Word Order (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| WORDORDER_01 | 0.9452 | 1.0000 | 1.0000 | 0.40 |
| WORDORDER_02 | 0.9591 | 1.0000 | 1.0000 | 0.42 |
| WORDORDER_03 | 0.9125 | 0.8831 | 0.8750 | 0.43 |

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| WORDORDER_04 | 0.9426 | 1.0000 | 1.0000 | 0.42 |
| WORDORDER_05 | 0.9039 | 0.8466 | 0.9219 | 0.44 |
| WORDORDER_06 | 0.9008 | 0.7799 | 0.8125 | 0.41 |

## C07: Related Topic (8 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| RELATED_01 | 0.8777 | 0.4576 | 0.7344 | 0.40 |
| RELATED_02 | 0.7925 | 0.3317 | 0.5938 | 0.38 |
| RELATED_03 | 0.8663 | 0.4503 | 0.6875 | 0.38 |
| RELATED_04 | 0.7887 | 0.2354 | 0.6250 | 0.35 |
| RELATED_05 | 0.8008 | 0.0962 | 0.5156 | 0.37 |
| RELATED_06 | 0.8392 | 0.1066 | 0.5469 | 0.36 |
| RELATED_07 | 0.8136 | 0.5056 | 0.7969 | 0.40 |
| RELATED_08 | 0.8370 | 0.1351 | 0.5313 | 0.38 |

## C08: Different Topic (10 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| DIFFTOPIC_01 | 0.7696 | 0.0570 | 0.6719 | 0.15 |
| DIFFTOPIC_02 | 0.8087 | 0.0000 | 0.5781 | 0.12 |
| DIFFTOPIC_03 | 0.8186 | 0.0442 | 0.5313 | 0.10 |
| DIFFTOPIC_04 | 0.8080 | 0.0000 | 0.4844 | 0.08 |
| DIFFTOPIC_05 | 0.7831 | 0.1156 | 0.6094 | 0.08 |
| DIFFTOPIC_06 | 0.8212 | 0.1113 | 0.6094 | 0.07 |
| DIFFTOPIC_07 | 0.7400 | 0.0570 | 0.4844 | 0.07 |
| DIFFTOPIC_08 | 0.8284 | 0.0000 | 0.5469 | 0.07 |
| DIFFTOPIC_09 | 0.7870 | 0.0000 | 0.5625 | 0.07 |
| DIFFTOPIC_10 | 0.8083 | 0.2030 | 0.6719 | 0.08 |

## C09: Short Text (8 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| SHORT_01 | 0.6894 | 0.0000 | 0.4219 | 0.20 |
| SHORT_02 | 0.4136 | 0.0000 | 0.4688 | 0.10 |
| SHORT_03 | 0.4158 | 0.0000 | 0.4375 | 0.15 |
| SHORT_04 | 0.4566 | 0.0000 | 0.3906 | 0.22 |
| SHORT_05 | 0.4225 | 0.0000 | 0.4531 | 0.10 |
| SHORT_06 | 0.6075 | 0.0000 | 0.6563 | 0.22 |

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| SHORT_07 | 0.7840 | 0.3361 | 0.8281 | 0.18 |
| SHORT_08 | 0.5097 | 0.0000 | 0.6563 | 0.20 |

## C10: Code vs Code (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| CODE_01 | 0.7178 | 0.5031 | 0.6719 | 0.75 |
| CODE_02 | 0.9310 | 1.0000 | 0.6563 | 0.88 |
| CODE_03 | 0.8501 | 0.4316 | 0.7344 | 0.72 |
| CODE_04 | 0.9225 | 0.6694 | 0.7969 | 0.70 |
| CODE_05 | 0.8974 | 0.8955 | 0.8594 | 0.82 |
| CODE_06 | 0.7954 | 0.2485 | 0.5938 | 0.58 |

## C11: Code vs Prose (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| CODEPROSE_01 | 0.7599 | 0.1278 | 0.5469 | 0.20 |
| CODEPROSE_02 | 0.7453 | 0.0644 | 0.6719 | 0.20 |
| CODEPROSE_03 | 0.6965 | 0.0522 | 0.5313 | 0.20 |
| CODEPROSE_04 | 0.7334 | 0.2202 | 0.4219 | 0.22 |
| CODEPROSE_05 | 0.7863 | 0.1130 | 0.5313 | 0.22 |
| CODEPROSE_06 | 0.7199 | 0.0000 | 0.4375 | 0.20 |

## C12: Cross-Lingual (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| CROSSLINGUAL_01 | 0.7479 | 0.0000 | 0.5781 | 0.10 |
| CROSSLINGUAL_02 | 0.8366 | 0.0000 | 0.4063 | 0.10 |
| CROSSLINGUAL_03 | 0.7967 | 0.0000 | 0.5469 | 0.10 |
| CROSSLINGUAL_04 | 0.7644 | 0.0000 | 0.5313 | 0.10 |
| CROSSLINGUAL_05 | 0.8585 | 0.0724 | 0.6875 | 0.10 |
| CROSSLINGUAL_06 | 0.7584 | 0.0000 | 0.4375 | 0.10 |

## C13: Noise (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| NOISE_01 | 0.6813 | 0.0000 | 0.5469 | 0.05 |
| NOISE_02 | 0.5905 | 0.0000 | 0.5156 | 0.05 |
| NOISE_03 | 0.6914 | 0.0000 | 0.4219 | 0.05 |
| NOISE_04 | 0.8878 | 0.0000 | 0.5000 | 0.05 |

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| NOISE_05 | 0.6833 | 0.0000 | 0.5313 | 0.05 |
| NOISE_06 | 0.5757 | 0.0000 | 0.5156 | 0.05 |

### C14: Length Diff (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| LENGTHDIFF_01 | 0.6853 | 0.2789 | 0.4844 | 0.60 |
| LENGTHDIFF_02 | 0.6475 | 0.2440 | 0.5938 | 0.58 |
| LENGTHDIFF_03 | 0.6573 | 0.2440 | 0.5938 | 0.58 |
| LENGTHDIFF_04 | 0.5954 | 0.2440 | 0.6094 | 0.57 |
| LENGTHDIFF_05 | 0.6910 | 0.4103 | 0.7500 | 0.55 |
| LENGTHDIFF_06 | 0.6642 | 0.2195 | 0.6875 | 0.58 |

### C15: Negation (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| NEGATION_01 | 0.9026 | 0.5884 | 0.8438 | 0.25 |
| NEGATION_02 | 0.8372 | 0.4111 | 0.5781 | 0.25 |
| NEGATION_03 | 0.9094 | 0.5056 | 0.7500 | 0.25 |
| NEGATION_04 | 0.8269 | 0.3808 | 0.5625 | 0.25 |
| NEGATION_05 | 0.8937 | 0.5727 | 0.8438 | 0.25 |
| NEGATION_06 | 0.9098 | 0.6694 | 0.8438 | 0.28 |

### C16: Factual Variation (6 pairs)

| Test ID | CHIMERA | TF-IDF | SimHash | GT |
|---|---|---|---|---|
| NUMVAR_01 | 0.9649 | 0.7523 | 0.8750 | 0.65 |
| NUMVAR_02 | 0.8474 | 0.6201 | 0.7813 | 0.62 |
| NUMVAR_03 | 0.9800 | 0.7523 | 0.8906 | 0.68 |
| NUMVAR_04 | 0.9457 | 0.7327 | 0.8281 | 0.62 |
| NUMVAR_05 | 0.8912 | 0.6392 | 0.7344 | 0.65 |
| NUMVAR_06 | 0.9651 | 0.6694 | 0.9219 | 0.63 |

## 5.6 Attractor Convergence Analysis

A structural property of CHIMERA-Hash is that all input texts converge to STABLE_ATTRACTOR at Level 5 regardless of content type. We verify this across four representative inputs spanning different domains.

| Text Type | L1 Mean | L2 Mean | L3 Mean | L4 Mean | L5 Mean | L5 Stab | State |
|---|---|---|---|---|---|---|---|
| "hello" | 106.40 | 105.80 | 104.40 | 102.40 | 103.60 | 0.9983 | STABLE_ATTRACTOR |
| English pangram | 94.35 | 94.69 | 93.12 | 88.19 | 96.81 | 0.9997 | STABLE_ATTRACTOR |
| SQL query | 71.19 | 78.31 | 85.88 | 93.12 | 93.94 | 0.9996 | STABLE_ATTRACTOR |
| Random noise | 77.92 | 75.33 | 81.17 | 86.42 | 93.42 | 0.9978 | STABLE_ATTRACTOR |

All four text types — short word, natural language, SQL code, and random noise — converge to STABLE_ATTRACTOR at exactly Level 5. The mean values approach the global attractor target (96.0) monotonically or near-monotonically, confirming the drift formula's convergence guarantee. This structural property — that every text, regardless of content, eventually reaches the same attractor basin — is unique to the CHIMERA family and has no equivalent in any competitor algorithm.

### 5.7 Speed and Hardware Analysis

| Algorithm | Mean ms/pair | Std ms | Mode | Hardware Notes |
|---|---|---|---|---|
| CHIMERA-Hash (full) | 3.7 | 1.8 | 4-component pipeline | i5 12th gen |
| CHIMERA-Hash (fast) | 0.82 | 0.1 | SRP only | i5 12th gen |
| TF-IDF | 1.4 | 0.7 | Vectorise + cosine | i5 12th gen |
| SimHash | 0.1 | 0.03 | Word token hash | i5 12th gen |
| MinHash | 0.1 | 0.02 | 3-gram shingles | i5 12th gen |

CHIMERA-Hash full pipeline requires 3.7 ms per pair on the test hardware. A fast mode (SimHash-SRP only) runs at 0.82 ms per pair and is suitable for large-scale pre-filtering before full evaluation. No GPU, external API, or pretrained model is required at any stage.

## 6. Discussion

### 6.1 Where CHIMERA-Hash Excels and Why

CHIMERA-Hash achieves its strongest results on near-duplicate detection (MAE 0.0109) and paraphrase identification (MAE 0.0755). These are precisely the categories where TF-IDF fails most severely (MAE 0.2112 and 0.6009 respectively). The reason is structural: TF-IDF can only detect similarity through shared vocabulary tokens. Two near-duplicates differing by one word ('jumped' vs 'jumps') share all but one token and receive high TF-IDF similarity. But two paraphrases ('automobile moving at high speed' vs 'car driving fast') may share no content tokens at all, yielding TF-IDF similarity of zero. CHIMERA-Hash, operating at the character level through the multiresolution encoder, detects that 'automobile' and 'car' share characters 'a', 'r' and are both four-to-ten letter strings, while the attractor trajectory detects that both texts evolve similarly from their respective initial means. The Chaos-IDF mechanism assigns character combinations discriminative weights that naturally group semantically related character patterns.

CHIMERA-Hash also handles length differences well (MAE 0.0754 vs TF-IDF 0.3140). When one text is a short phrase and the other is a full sentence on the same topic, TF-IDF's cosine normalisation over the full vocabulary space penalises the length difference. CHIMERA-Hash's character-level encoding with chaos-IDF weighting naturally handles variable-length inputs because importance is determined per character-position pair, not normalised by document length.

## 6.2 Where TF-IDF Excels and Why

TF-IDF wins decisively on topic discrimination (Different Topic: MAE 0.0702 vs CHIMERA 0.7083), cross-lingual pairs (0.0879 vs 0.6938), and noise rejection (0.0500 vs 0.6349). The common thread: all three categories require detecting that two texts share essentially no content, and vocabulary overlap is the most reliable signal for this. When 'the stock market' appears in text A and 'deep sea fish' in text B, a TF-IDF model immediately assigns near-zero similarity because no tokens are shared. CHIMERA-Hash, operating at the character level, finds that both are medium-length English prose with similar character distributions and assigns moderate similarity regardless.

This is a fundamental limitation of character-level and structural encodings: character distributions are largely determined by the language of the text, not its topic. All English prose shares a characteristic distribution over the 26 letters. Only a semantic or vocabulary-level signal can reliably detect that 'stock market' and 'deep sea fish' are unrelated.

## 6.3 Recommended Hybrid Architecture

The pattern of results suggests a natural hybrid: use TF-IDF as a fast topic filter (document retrieval stage) and CHIMERA-Hash as a fine-grained similarity scorer (reranking stage). In a plagiarism detection pipeline: (1) retrieve candidate documents using TF-IDF or BM25 top-k retrieval, eliminating completely unrelated documents; (2) apply CHIMERA-Hash to the retrieved candidates for fine-grained near-duplicate and paraphrase detection. This combines TF-IDF's strength at topic discrimination with CHIMERA-Hash's strength at detecting paraphrase and structural similarity within topically related documents.

## 6.4 Limitations

CHIMERA-Hash has four identified limitations. First, it does not perform semantic reasoning: it cannot detect that 'the patient recovered' and 'the patient did not recover' are opposite in meaning — both receive high similarity because they share nearly all characters. The negation category (MAE 0.6249) illustrates this clearly. Second, CHIMERA-Hash is language-agnostic in the wrong direction: it cannot detect that 'the weather is beautiful' and 'Le chat mange du poisson' are linguistically unrelated, only that they are both prose strings of similar length. Third, the component weights (0.40 / 0.30 / 0.20 / 0.10) are manually tuned and not optimised on a labelled training set. Fourth, performance on very short texts (one to three words) is unpredictable because the multiresolution encoder has few features to extract.

## 6.5 Evolution of the Attractor Family

| Ver | Name | Correlation | Key Addition | Outcome |
|---|---|---|---|---|
| v1 | DSF-V3 | 0.5438 (16-pair) | Logistic map attractor trajectory | Established attractor baseline |
| v2 | FlyHash | 0.5538 (16-pair) | Biological SDR + holographic encoding | Marginal improvement |

| Ver | Name | Correlation | Key Addition | Outcome |
|---|---|---|---|---|
| v3 | GAQH | 0.3274 (16-pair) | Quantum superposition matrices | Severe regression — all texts oversimilar |
| v4 | CHIMERA-Hash | 0.5082 (115-pair) | Chaos-IDF + SimHash-SRP + MR encoder | Honest 2nd place; 1st on 6 categories |

The progression shows that complexity does not guarantee improvement. GAQH's quantum framing added unnecessary complexity and caused a regression. CHIMERA-Hash returns to mathematically grounded components (CountSketch, SRP, logistic map) while introducing two genuinely novel mechanisms. The result is a more honest benchmark that uses 115 pairs rather than 16, revealing the true performance profile more clearly.

# 7. Conclusion

We introduced CHIMERA-Hash, a text fingerprinting algorithm combining Count Sketch, SimHash-SRP, multiresolution character encoding, and chaotic attractor trajectory into a unified similarity function. On a 115-pair benchmark spanning 16 challenge categories, CHIMERA-Hash ranks second overall by Pearson correlation (0.5082) after TF-IDF (0.5683), and first on six categories including near-duplicate detection (MAE 0.0109 vs TF-IDF 0.2112) and paraphrase identification (MAE 0.0755 vs TF-IDF 0.6009).

The algorithm introduces two mechanisms with no identified prior art: Chaos-IDF weighting, which replaces corpus-dependent IDF with logistic map sensitivity at r = 3.9, enabling corpus-free term importance discrimination; and LSH-to-attractor seeding, which couples SimHash bit-signatures to chaotic trajectory initialisation, creating mutual reinforcement between the static angular hash and the dynamic structural fingerprint. The chaotic attractor converges to STABLE_ATTRACTOR at Level 5 for all text types tested. The full algorithm requires 3.7 ms per pair on a laptop CPU with no GPU.

Future directions include: (1) combining CHIMERA-Hash with a TF-IDF topic filter in a two-stage pipeline to cover topic discrimination and cross-lingual cases; (2) learning component weights on a labelled similarity dataset (e.g., STS-Benchmark or MRPC) to replace manually tuned values; (3) extending Chaos-IDF to word-level n-gram features for better semantic coverage; (4) investigating the theoretical relationship between the Chaos-IDF importance distribution and standard IDF as a function of logistic map parameter r; and (5) applying CHIMERA-Hash to code clone detection, where character-level structural similarity is more informative than vocabulary overlap.

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# Appendix A: Full Benchmark — All 115 Text Pairs

All test pairs, texts, and ground-truth scores used in Section 5. Scores computed live.

| ID | Text A | Text B | GT |
|---|---|---|---|
| 01 | The quick brown fox jumps over the lazy dog | The quick brown fox jumps over the lazy dog | 1.00 |
| 02 | Machine learning is a subset of artificial intelligence | Machine learning is a subset of artificial intelligence | 1.00 |
| 03 | def fibonacci(n): return n if n<=1 else fibonacci(n-1)+fibonacci(n-2) | def fibonacci(n): return n if n<=1 else fibonacci(n-1)+fibonacci(n-2) | 1.00 |
| 04 | The stock market closed higher today driven by technology stocks | The stock market closed higher today driven by technology stocks | 1.00 |
| 05 | climate change global warming temperature rise | climate change global warming temperature rise | 1.00 |
| 01 | The quick brown fox jumps over the lazy dog | The quick brown fox jumped over the lazy dog | 0.95 |
| 02 | Machine learning is transforming the technology industry rapidly | Machine learning is transforming technology industries rapidly | 0.93 |
| 03 | Scientists have discovered a new planet orbiting a distant star | Scientists discovered a new planet orbiting a distant star system | 0.94 |
| 04 | The president signed the new economic policy into law yesterday | The president signed the new economic policy into law on Thursday | 0.93 |
| 05 | Python is widely used for data science and machine learning tasks | Python is widely used for data science and machine learning applications | 0.93 |
| 06 | The hospital reported a significant decrease in patient wait times | The hospital reported a significant reduction in patient waiting times | 0.92 |
| 07 | Researchers published a groundbreaking study on cancer treatment | Researchers have published a groundbreaking study on cancer treatments | 0.93 |
| 08 | The electric vehicle market grew by forty percent last quarter | The electric vehicle market grew by 40 percent in the last quarter | 0.92 |
| 09 | Apple released its latest smartphone with improved camera features | Apple released its newest smartphone with improved camera capabilities | 0.92 |
| 10 | The new law requires companies to disclose their carbon emissions annually | The new regulation requires companies to report their carbon emissions every ... | 0.90 |
| 01 | The automobile moved at high speed down the highway | The car was driving fast on the road | 0.75 |
| 02 | Artificial intelligence will revolutionize how humans work | AI systems are going to completely change the nature of employment | 0.72 |
| 03 | The child was very happy when she received the gift | The little girl felt extremely joyful upon getting the present | 0.73 |
| 04 | The company reported record profits for the financial year | The firm announced its highest ever earnings during the fiscal period | 0.72 |
| 05 | Doctors recommend exercising regularly to maintain good health | Physicians advise people to work out frequently for better wellbeing | 0.72 |

| ID | Text A | Text B | GT |
|---|---|---|---|
| 06 | The storm caused widespread damage across the coastal region | Heavy winds and rain devastated communities along the shoreline | 0.70 |
| 07 | The government announced new measures to reduce unemployment | Officials unveiled fresh policies aimed at tackling joblessness | 0.71 |
| 08 | Students who study consistently tend to perform better in exams | Pupils who revise regularly usually achieve higher test scores | 0.72 |
| 09 | The film received critical acclaim for its stunning cinematography | Critics praised the movie for its breathtaking visual storytelling | 0.73 |
| 10 | Eating a balanced diet is essential for long term health | Consuming nutritious food regularly is vital for your wellbeing over time | 0.71 |
| 11 | The software update introduced several new security patches | The latest version of the program included multiple fixes for vulnerabilities | 0.70 |
| 12 | Global temperatures have risen significantly over the past century | Average worldwide heat levels have increased substantially during the last hu... | 0.71 |
| 01 | Climate change is causing more frequent and severe weather events worldwide | Global warming has led to an increase in the frequency and intensity of extre... | 0.70 |
| 02 | The neural network achieved state of the art performance on the benchmark | Our deep learning model set new records across all evaluation metrics on the ... | 0.68 |
| 03 | The central bank raised interest rates to combat rising inflation | Monetary authorities increased borrowing costs in an effort to control accele... | 0.68 |
| 04 | Renewable energy sources are becoming increasingly cost competitive | The price of clean power generation has fallen dramatically making it economi... | 0.67 |
| 05 | The quarterback threw a touchdown pass in the final seconds | The football player scored the winning points with a last second throw to his... | 0.66 |
| 06 | The algorithm processes millions of records in under one second | This computational method can handle vast datasets with extremely low latency | 0.67 |
| 07 | The prime minister resigned following the no confidence vote | The head of government stepped down after lawmakers withdrew their support | 0.67 |
| 08 | Researchers found a strong correlation between sleep quality and cognitive pe... | Scientists discovered that better rest is linked to improved mental functioni... | 0.68 |
| 01 | The experimental results demonstrate a statistically significant correlation. | Our findings indicate a measurable relationship between the observed parameters. | 0.65 |
| 02 | Pursuant to the terms outlined in Section 4.2 of the agreement herein. | In accordance with the provisions specified under Clause 7.1 of this contract. | 0.63 |
| 03 | We hereby acknowledge receipt of your communication dated the first instant. | This letter confirms that we have received your correspondence of recent date. | 0.62 |
| 04 | The patient presented with acute onset chest pain and shortness of breath. | The subject exhibited sudden thoracic discomfort accompanied by respiratory d... | 0.64 |
| 05 | Yo bro that game last night was absolutely insane no cap fr. | Dude that match was totally wild yesterday I am not even joking. | 0.63 |
| 06 | The Board of Directors resolved to approve the proposed merger subject to reg... | Trustees voted to sanction the acquisition pending approval from the relevant... | 0.62 |
| 01 | The dog bit the man | The man bit the dog | 0.40 |

| ID | Text A | Text B | GT |
|---|---|---|---|
| 02 | John loves Mary deeply and sincerely | Mary loves John deeply and sincerely | 0.42 |
| 03 | The police arrested the suspect near the bank | The suspect was arrested near the bank by the police | 0.43 |
| 04 | Cats eat mice and dogs chase cats | Dogs chase cats and cats eat mice | 0.42 |
| 05 | The teacher praised the student for excellent work | The student was praised by the teacher for excellent work | 0.44 |
| 06 | Fast cars use more fuel than slow ones | Slow cars use less fuel than fast ones | 0.41 |
| 01 | Python is a programming language used for data science | Java is a programming language used for enterprise applications | 0.40 |
| 02 | The iPhone was released by Apple in 2007 | The Samsung Galaxy was released by Samsung to compete with Apple | 0.38 |
| 03 | Football is the most popular sport in the United States | Basketball is one of the fastest growing sports in the United States | 0.38 |
| 04 | World War One began in 1914 in Europe | World War Two ended in 1945 with the defeat of Nazi Germany | 0.35 |
| 05 | NASA launched the Artemis mission to return humans to the Moon | SpaceX developed the Starship rocket for Mars colonization missions | 0.37 |
| 06 | The stock market rally was driven by strong earnings reports | The bond market declined as investors moved to higher yielding assets | 0.36 |
| 07 | Python uses indentation to define code blocks | JavaScript uses curly braces to define code blocks | 0.40 |
| 08 | Convolutional neural networks excel at image recognition tasks | Recurrent neural networks are designed for sequential data processing | 0.38 |
| 01 | The stock market experienced significant volatility during trading | Scientists discovered a new species of deep sea fish near the trench | 0.15 |
| 02 | Python is a high level programming language known for readability | The Amazon rainforest contains half of all species on the planet | 0.12 |
| 03 | The recipe calls for two cups of flour and one egg | The space shuttle was launched from Kennedy Space Center in Florida | 0.10 |
| 04 | Manchester United won the Premier League championship last season | Quantum computers use qubits instead of classical binary bits | 0.08 |
| 05 | The Eiffel Tower was built in Paris in 1889 | Antibiotics are used to treat bacterial infections in medicine | 0.08 |
| 06 | The violin is a string instrument played with a bow | The national debt of the United States exceeds thirty trillion dollars | 0.07 |
| 07 | Photosynthesis converts sunlight into glucose in plant cells | The Roman Empire fell in 476 AD after centuries of decline | 0.07 |
| 08 | The treaty was signed by representatives of fifteen nations | Baking bread requires yeast flour water and salt mixed together | 0.07 |
| 09 | The jaguar is the largest wild cat in the Americas | Object oriented programming uses classes and inheritance structures | 0.07 |
| 10 | The Himalayan mountain range contains the world's highest peaks | The Federal Reserve sets monetary policy for the United States economy | 0.08 |
| 01 | hello | world | 0.20 |
| 02 | hi | bye | 0.10 |
| 03 | cat | dog | 0.15 |

| ID | Text A | Text B | GT |
|---|---|---|---|
| 04 | good | great | 0.22 |
| 05 | yes | no | 0.10 |
| 06 | fast car | slow truck | 0.22 |
| 07 | I love pizza | I hate pizza | 0.18 |
| 08 | run fast | walk slowly | 0.20 |
| 01 | def add(a, b): return a + b | def sum_two(x, y): return x + y | 0.75 |
| 02 | for i in range(10): print(i) | for j in range(10): print(j) | 0.88 |
| 03 | SELECT * FROM users WHERE id = 1 | SELECT * FROM customers WHERE customer_id = 1 | 0.72 |
| 04 | import numpy as np arr = np.zeros(100) | import numpy as np array = np.ones(100) | 0.70 |
| 05 | if x > 0: return True else: return False | if value > 0: return True else: return False | 0.82 |
| 06 | def factorial(n): return 1 if n==0 else n*factorial(n-1) | def fib(n): return n if n<=1 else fib(n-1)+fib(n-2) | 0.58 |
| 01 | def fibonacci(n): return n if n<=1 else fibonacci(n-1)+fibonacci(n-2) | The Fibonacci sequence is a series where each number equals the sum of two pr... | 0.20 |
| 02 | SELECT name FROM employees WHERE salary > 50000 | Find all employees whose annual pay exceeds fifty thousand dollars | 0.20 |
| 03 | for i in range(len(arr)): arr[i] = arr[i] * 2 | Multiply every element in the array by two | 0.20 |
| 04 | class Animal: def __init__(self, name): self.name = name | An animal object stores a name attribute when created | 0.22 |
| 05 | import pandas as pd df = pd.read_csv('data.csv') | Load a CSV file into a dataframe using the pandas library | 0.22 |
| 06 | try: x = int(input()) except ValueError: print('invalid') | If the user enters something that is not a number show an error message | 0.20 |
| 01 | The weather is beautiful today with sunshine everywhere | Le chat mange du poisson dans la cuisine ce soir | 0.10 |
| 02 | Machine learning algorithms transform data analysis | Los algoritmos de aprendizaje automatico transforman el analisis de datos | 0.10 |
| 03 | The economy is growing at a steady pace | Die Wirtschaft wachst mit einem stetigen Tempo | 0.10 |
| 04 | Scientists discovered a cure for the disease | Gli scienziati hanno scoperto una cura per la malattia | 0.10 |
| 05 | The president made an important announcement today | Le president a fait une annonce importante aujourd hui | 0.10 |
| 06 | Children love playing games in the park | Los ninos aman jugar juegos en el parque por la tarde | 0.10 |
| 01 | The weather today is sunny with mild temperatures | xK92mPvL8nQ3rT7wY1oE5uI6hG4jF0dA | 0.05 |
| 02 | Machine learning is a branch of artificial intelligence | 7hJ#mK9$pL2@nQ5^rT8&wX1!yZ4*bC6 | 0.05 |
| 03 | The capital of France is Paris | aBcDeFgHiJkLmNoPqRsTuVwXyZ012345 | 0.05 |
| 04 | asdfjkl qwerty uiop zxcvbnm | poiuytrewq lkjhgfdsa mnbvcxz | 0.05 |
| 05 | 1234567890 abcdef ghijk | 9876543210 zyxwvu tsrqp | 0.05 |

| ID | Text A | Text B | GT |
|---|---|---|---|
| 06 | The results show a clear improvement | !@#$%^&*()_+{}\|:<>?~`-=[]; ',./ | 0.05 |
| 01 | Python programming | Python is a high level interpreted programming language with dynamic semantic... | 0.60 |
| 02 | Climate change | Climate change refers to long term shifts in global temperatures and weather ... | 0.58 |
| 03 | Neural network | A neural network is a computational model inspired by the human brain consist... | 0.58 |
| 04 | Stock market | The stock market is a platform where buyers and sellers trade shares of publi... | 0.57 |
| 05 | Quantum computing | Quantum computing leverages quantum mechanical phenomena such as superpositio... | 0.55 |
| 06 | Machine learning | Machine learning is a method of data analysis that automates analytical model... | 0.58 |
| 01 | The patient recovered fully after the surgery | The patient did not recover after the surgery | 0.25 |
| 02 | The economy is growing at a record pace | The economy is not growing and is in recession | 0.25 |
| 03 | The algorithm converges quickly on large datasets | The algorithm fails to converge on large datasets | 0.25 |
| 04 | Smoking is harmful to your health | Smoking is not harmful and has no health effects | 0.25 |
| 05 | The company made a profit last quarter | The company did not make a profit last quarter | 0.25 |
| 06 | The test results were positive | The test results were negative | 0.28 |
| 01 | The population of India is 1.4 billion people | The population of China is 1.4 billion people | 0.65 |
| 02 | The company was founded in 1998 in California | The company was founded in 2004 in New York | 0.62 |
| 03 | The temperature today is 25 degrees Celsius | The temperature today is 35 degrees Celsius | 0.68 |
| 04 | The train arrives at platform 3 at 10 AM | The train arrives at platform 7 at 2 PM | 0.62 |
| 05 | Revenue grew by 15 percent in the third quarter | Revenue grew by 32 percent in the second quarter | 0.65 |
| 06 | The building has 50 floors and was completed in 2010 | The building has 80 floors and was completed in 2018 | 0.63 |

# Appendix B: Public Links

| Resource | URL |
|---|---|
| Source Code | https://github.com/nickzq7/chimera-hash |
| YouTube Channel | https://www.youtube.com/@ProgramDr |

| Resource | URL |
|---|---|
| LinkedIn | https://www.linkedin.com/in/manish-parihar-899b5b23a/ |
| Benchmark Script | github.com/nickzq7/chimera-hash/blob/main/run_benchmark.py |

# Appendix C: Reproducibility

All experiments can be reproduced with:

**pip install numpy scikit-learn**

**python run_benchmark.py**

Expected output: CHIMERA-Hash ranks #2 overall with correlation 0.5082, first place on Near-Duplicate (MAE 0.0109) and Paraphrase (MAE 0.0755) categories. Hardware: any machine with Python 3.8+ and 4 GB RAM. GPU not required. Runtime: approximately 3 minutes for all 115 pairs x 4 algorithms.